

COMPENSATION RELATIONSHIP IN THERMODYNAMICS OF GAS-PHASE MOLECULAR COMPLEXATION

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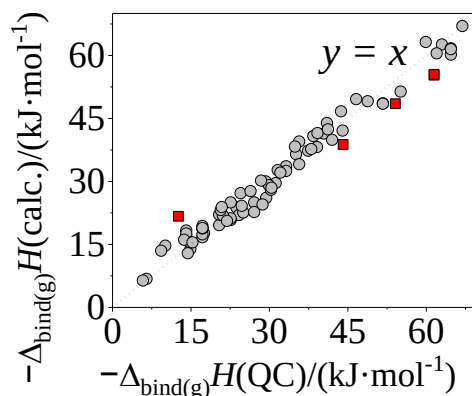
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The phenomenon of compensation between changes in enthalpy and entropy in various physicochemical processes has been a subject of interest for scientists for over 90 years. In previous works, linear relationships were obtained between the changes in Gibbs energies and enthalpies for solvation and solution-phase complexation processes, which share the same slope [1].

The common slope in these dependences allows deriving a similar equation for complexation in the gas phase:

$$\Delta_{\text{bind(g)}}G / (\text{kJ} \cdot \text{mol}^{-1}) = 0,660 \cdot \Delta_{\text{bind(g)}}H / (\text{kJ} \cdot \text{mol}^{-1}) + 17,0 + 2,5 \cdot n$$

In the present work, a comparison was made between the enthalpy values calculated using the obtained relationship from experimental Gibbs energy values and the values computed by quantum-chemical methods for 67 donor-acceptor pairs (graphical representation of this comparison is given in fig. below) [2].



The relationship obtained in present work allows creating an approach for determining binding enthalpies in the gas phase from a single value of the equilibrium constant at 298.15 K. Furthermore, the obtained results mean that, in each case, the compensation relationships have a common origin, which is beyond experimental artifacts, group additivity in homologous series, or solvent reorganization effects.

1. Solomonov B. N., Khislamiev M. B., Yagofarov M. I. // J. Mol. Liq. 2024. Vol. 411, Article ID 125690. <https://doi.org/10.1016/j.molliq.2024.125690>

2. Sokolov A.A., Khislamiev M.B., Solomonov B.N., Yagofarov M.I. // Phys. Chem. Chem. Phys. 2025. Vol. 27, P. 26139–26151. <https://doi.org/10.1039/D5CP04241K>

The study was supported by the Russian Science Foundation, project No. 25-73-20073.